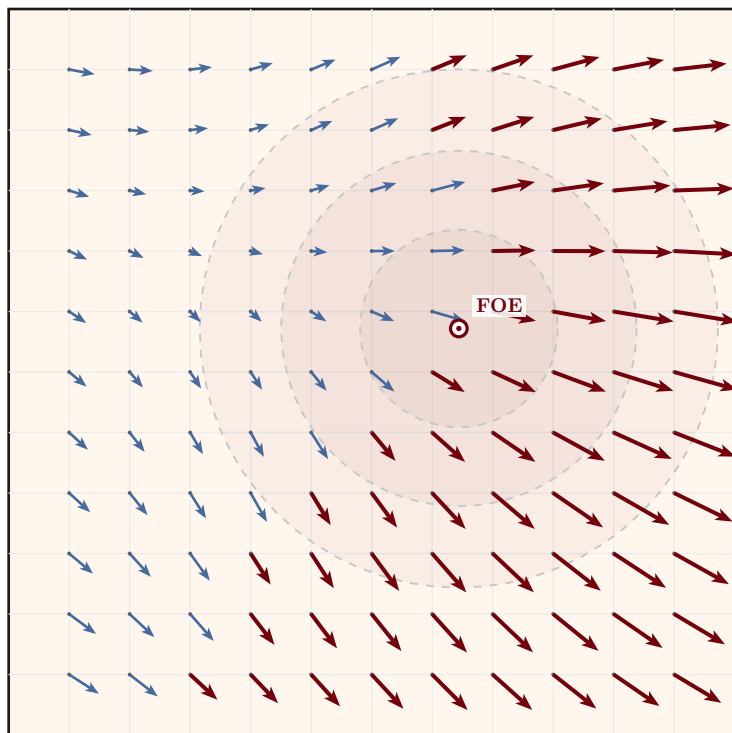


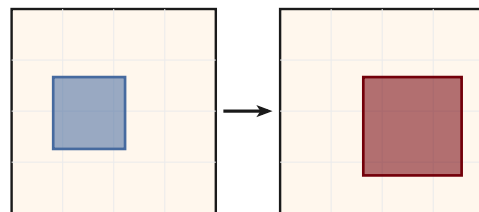
Optical flow field

dense motion $\mathbf{v}(\mathbf{x}) = (u, v)$ from $I_t \rightarrow I_{t+1}$, one arrow per pixel grid cell



$$\text{brightness constancy } I_{t(x)} = I_{t+1}(x + v) \rightarrow I_x u + I_y v + I_t = 0$$

input frames

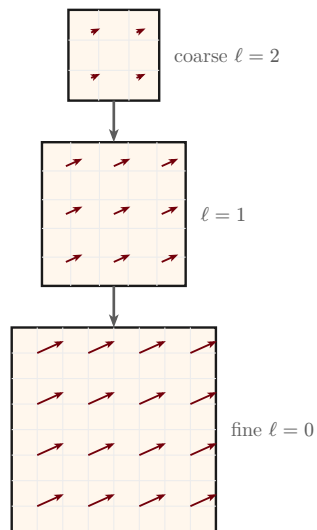


I_t

I_{t+1}

displacement of the patch $\rightarrow$ flow $\mathbf{v}$

coarse-to-fine warping



upsample $\times 2$ + warp $\rightarrow$ refine

→ fast — object (looming)

→ slow — background pan

⊙ focus of expansion (FOE)